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AI for Bharat Hackathon

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Problem Statement : Boliye-ji - Voice First AI assistant for Discovering Govenment Schemes

Brief about the Idea:

Many students miss government scholarships and benefits simply because they don't know they exist or don't know how to apply. The information is spread across many confusing websites, written in difficult language, and is hard to understand. Because of this, deserving students - especially those from rural or low-income backgrounds - often lose access to important support.

Boliye-ji is a voice-first AI assistant designed to help students easily discover and understand government schemes using simple spoken Hindi. Many eligible students miss out on scholarships and benefits because information is scattered across complex websites, written in bureaucratic language, and difficult to navigate. Boliye-ji removes these barriers by letting users simply speak.

The assistant asks a few guided questions about the student's education level, family income, and location, then automatically matches them to relevant government schemes using clear eligibility rules. Instead of showing long documents, it explains benefits, eligibility reasons, required documents, and application steps through natural voice responses.

The entire experience is hands-free, accessible, and designed for low digital literacy, making it especially useful for first-generation learners. Built on a scalable AWS-based architecture, Boliye-ji serves as a voice gateway between students and public services, ensuring that government benefits reach the people they are intended for.

- ## How different is it from any of the other existing ideas?

  Existing platforms list government schemes as text-heavy portals or generic chatbots that assume users know what to search for and can interpret eligibility rules on their own. **Boliye-ji** flips this model by starting with the user, using a voice-first, Hindi interaction to understand a student's profile and automatically match them to relevant schemes through deterministic eligibility logic. Instead of returning documents or links, it explains benefits, eligibility reasons, and next steps in simple spoken language, making access possible even for users with low digital literacy. This shifts government information from something users must navigate to something that actively reaches and guides them.

- ## How will it be able to solve the problem?

  Boliye-ji solves the access gap by converting complex, scattered government scheme information into a guided voice experience that requires no prior knowledge or technical skill. By asking a small set of simple questions, it accurately identifies which schemes a student is eligible for and filters out irrelevant information, reducing confusion and effort. The system then delivers clear, spoken explanations and actionable steps, allowing students to move from awareness to application in a single interaction. This approach directly addresses the core problem of discoverability and comprehension, ensuring that government benefits are understood and usable by the people they are intended for.

- ## USP of the proposed solution

  The main selling point is that Boliye-ji transforms complex government scheme information into a simple voice conversation, allowing anyone to discover and act on the benefits they are eligible for without reading, searching, or navigating digital portals. Picture being able to CALL a number, and instantly get responses regarding these schemes, seamlessly , without any prior knowledge.

List of features offered by the solution

- **Phone-call based access**
  Users can simply call a number; Boliye-ji answers and starts the conversation, removing the need for apps, logins, or smartphones.

- **Hindi voice-first interaction**
  Complete hands-free experience with spoken input and output, designed for users with low digital literacy.

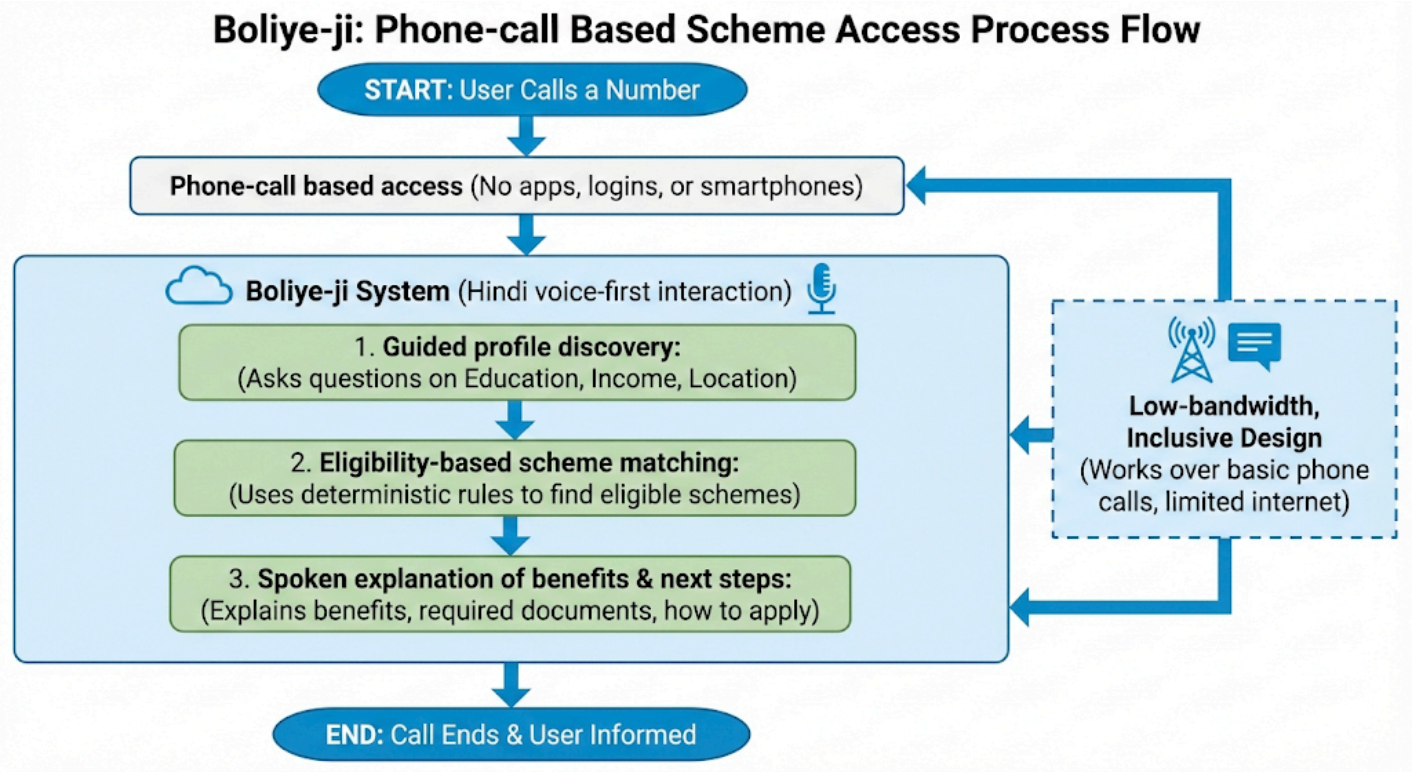
- **Guided profile discovery**
  The assistant asks a few simple questions to understand the student's education level, income range, and location.

- **Eligibility-based scheme matching**
  Matches the user only to schemes they are actually eligible for using deterministic rules, avoiding confusion from irrelevant results.

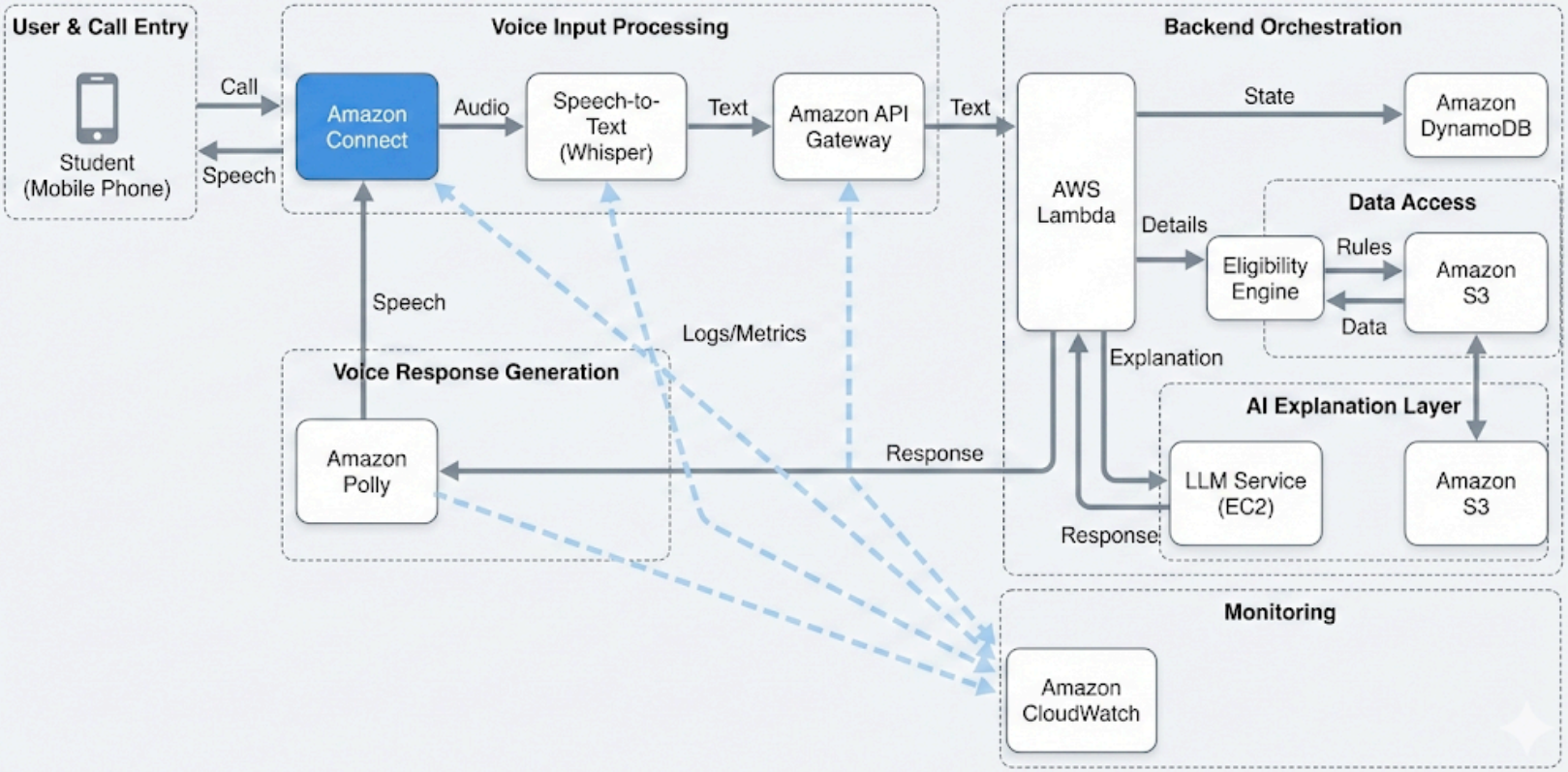
- **Spoken explanation of benefits and next steps**
  Clearly explains what the user gets, why they qualify, required documents, and how to apply.

- **Low-bandwidth, inclusive design**
  Works over basic phone calls and is usable even in areas with limited internet access.

Process flow diagram or Use-case diagram

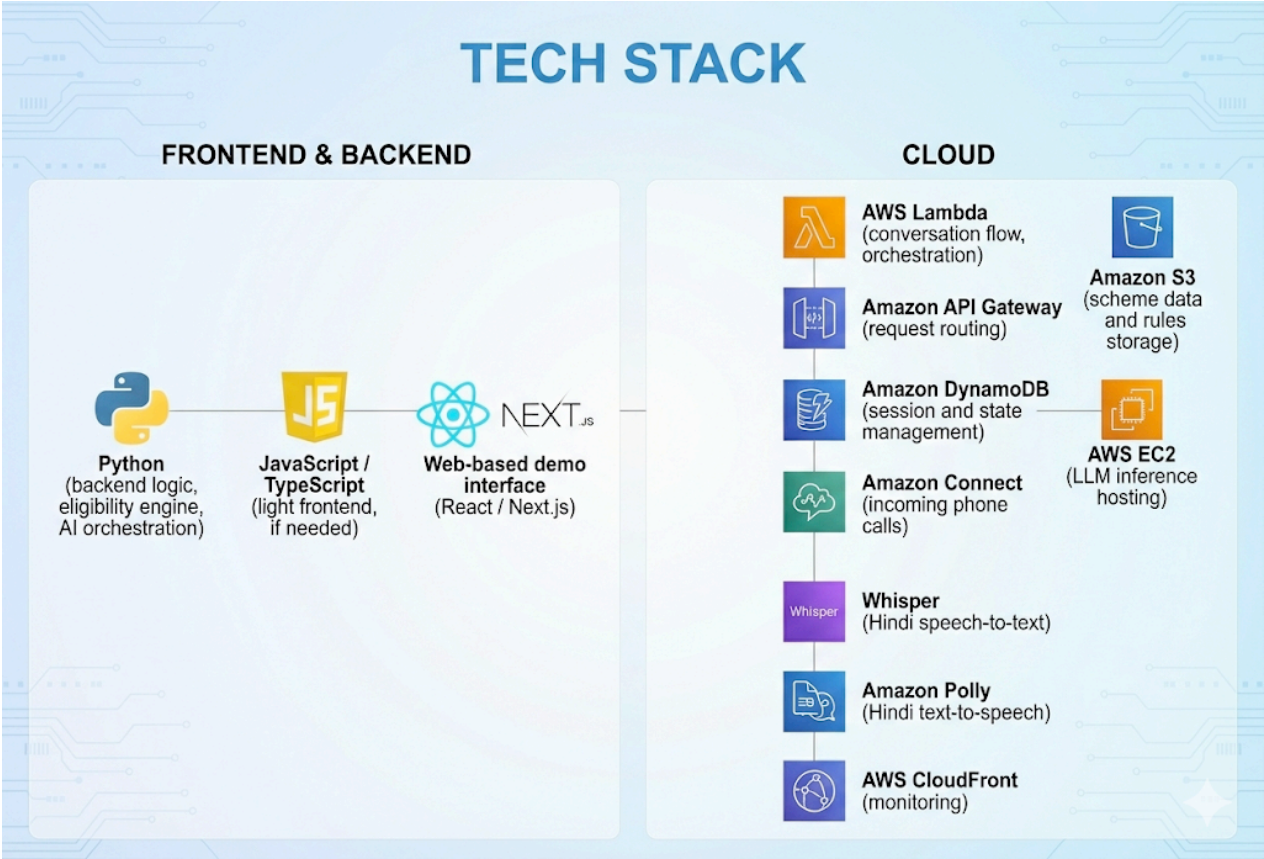


Boliye-ji Architecture Diagram: Voice-First AI Assistant



Technologies to be used in the solution:

- Python (backend logic, eligibility engine, AI orchestration)
- JavaScript / TypeScript (light frontend, if needed)
- Web-based demo interface (React / Next.js)
- AWS Lambda (conversation flow, orchestration)
- Amazon API Gateway (request routing)
- Amazon DynamoDB (session and state management)
- Amazon S3 (scheme data and rules storage)
- AWS EC2 (LLM inference hosting
- Amazon Connect (incoming phone calls)
- Whisper (Hindi speech-to-text)
- Amazon Polly (Hindi text-to-speech)



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Thank You